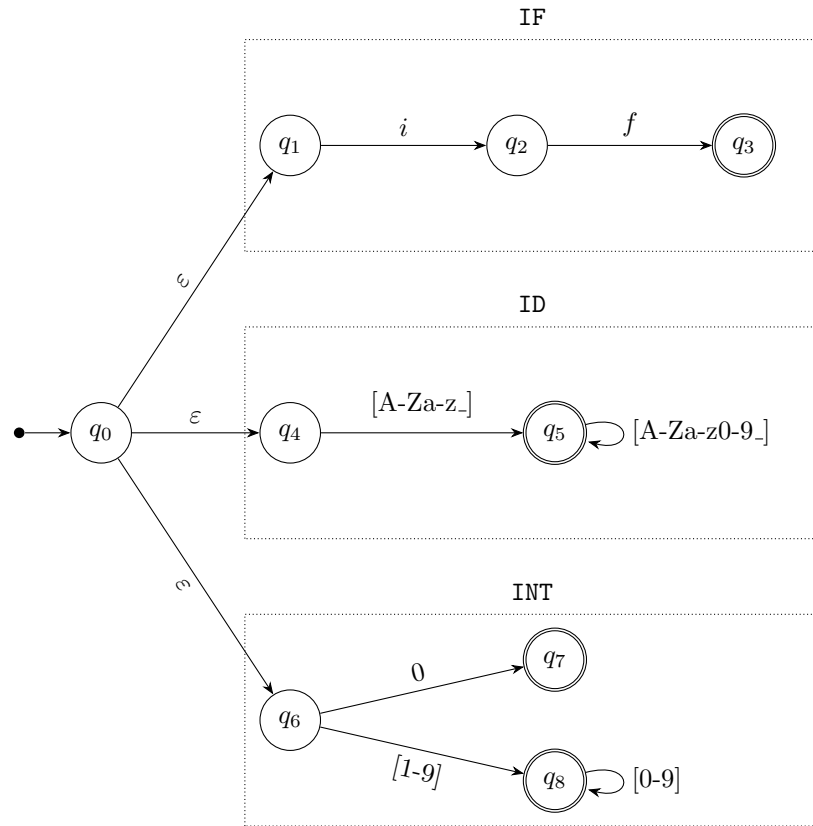


CSE411 Assignment #2: Applying the Subset Construction Algorithm to **IF** | **ID** | **INT**

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1 NFA



2 Step-by-Step Conversion to DFA

2.1 Identifying D0, D, W, and F_N

variable	value
N0	q_0
D0	$\varepsilon\text{-closure}(\{N0\}) = \varepsilon\text{-closure}(\{q_0\}) = \{q_0, q_1, q_4, q_6\}$
D	$\{D0\} = \{\{q_0, q_1, q_4, q_6\}\}$
W	$\{D0\} = \{\{q_0, q_1, q_4, q_6\}\}$
F_N	$\{q_3, q_5, q_7, q_8\}$

2.2 Intermediate Automata

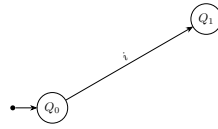
line	operation	value
1	$D0 := \varepsilon\text{-closure}(\{N0\})$	$D0 = Q_0 = \{q_0, q_1, q_4, q_6\}$
2	$D := \{D0\}$	$D = \{Q_0\}$



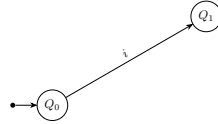
line	operation	value
3	$W := \{D0\}$	$W = \{Q_0\}$
4	while $W \neq \emptyset$	$W = \{Q_0\} \neq \emptyset$
5	remove Q from W	$Q = Q_0, W = \{\}$
6	for $\alpha \in \Sigma$	$\alpha = i$
7	$T := \emptyset$	$T = \emptyset$
8	for $S \in Q$	$S = q_0$
9	$T := T \cup \delta_N(S, \alpha)$	$\emptyset \rightarrow \emptyset$
8	for $S \in Q$	$S = q_1$
9	$T := T \cup \delta_N(S, \alpha)$	$\emptyset \rightarrow \{q_2\}$
8	for $S \in Q$	$S = q_4$
9	$T := T \cup \delta_N(S, \alpha)$	$\{q_2\} \rightarrow \{q_2, q_5\}$
8	for $S \in Q$	$S = q_6$
9	$T := T \cup \delta_N(S, \alpha)$	$\{q_2, q_5\} \rightarrow \{q_2, q_5\}$
10	$T' := \varepsilon\text{-closure}(T)$	$T' = \{q_2, q_5\} = Q_1$
11	$D' := D \cup \{T'\}$	$D' = \{Q_0, Q_1\}$



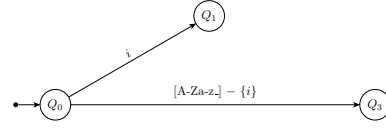
line	operation	value
12	$\delta_D(Q, \alpha) := T'$	$\delta_D(Q_0, \alpha) = Q_1$



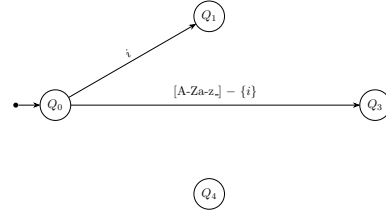
line	operation	value
13	if $T' \notin D$	$Q_1 \notin D$
14	$W := W \cup \{T'\}$	$W = \{Q_1\}$
15	$D := D'$	$D = \{Q_0, Q_1\}$
6	for $\alpha \in \Sigma$	$\alpha \in [A-Za-z] - \{i\}$
7	$T := \emptyset$	$T = \emptyset$
8	for $S \in Q$	$S = q_0$
9	$T := T \cup \delta_N(S, \alpha)$	$\emptyset \rightarrow \emptyset$
8	for $S \in Q$	$S = q_1$
9	$T := T \cup \delta_N(S, \alpha)$	$\emptyset \rightarrow \emptyset$
8	for $S \in Q$	$S = q_4$
9	$T := T \cup \delta_N(S, \alpha)$	$\emptyset \rightarrow \{q_5\}$
8	for $S \in Q$	$S = q_6$
9	$T := T \cup \delta_N(S, \alpha)$	$\{q_5\} \rightarrow \{q_5\}$
10	$T' := \varepsilon\text{-closure}(T)$	$T' = \{q_5\} = Q_3$
11	$D' := D \cup \{T'\}$	$D' = \{Q_0, Q_1, Q_3\}$



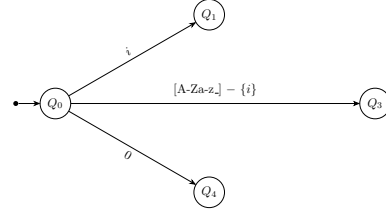
line	operation	value
12	$\delta_D(Q, \alpha) := T'$	$\delta_D(Q_0, \alpha) = Q_3$



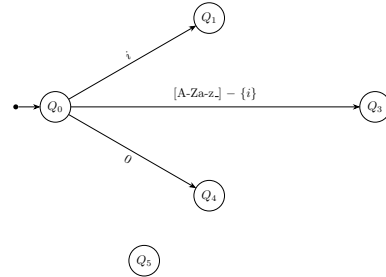
line	operation	value
13	if $T' \notin D$	$Q_3 \notin D$
14	$W := W \cup \{T'\}$	$W = \{Q_1, Q_3\}$
15	$D := D'$	$D = \{Q_0, Q_1, Q_3\}$
6	for $\alpha \in \Sigma$	$\alpha = 0$
7	$T := \emptyset$	$T = \emptyset$
8	for $S \in Q$	$S = q_0$
9	$T := T \cup \delta_N(S, \alpha)$	$\emptyset \rightarrow \emptyset$
8	for $S \in Q$	$S = q_1$
9	$T := T \cup \delta_N(S, \alpha)$	$\emptyset \rightarrow \emptyset$
8	for $S \in Q$	$S = q_4$
9	$T := T \cup \delta_N(S, \alpha)$	$\emptyset \rightarrow \emptyset$
8	for $S \in Q$	$S = q_6$
9	$T := T \cup \delta_N(S, \alpha)$	$\emptyset \rightarrow \{q_7\}$
10	$T' := \varepsilon\text{-closure}(T)$	$T' = \{q_7\} = Q_4$
11	$D' := D \cup \{T'\}$	$D' = \{Q_0, Q_1, Q_3, Q_4\}$



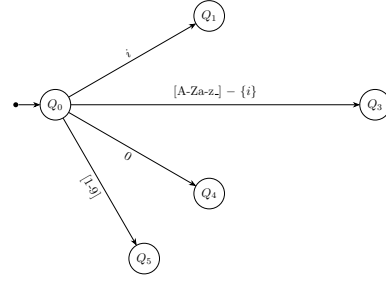
line	operation	value
12	$\delta_D(Q, \alpha) := T'$	$\delta_D(Q_0, \alpha) = Q_4$



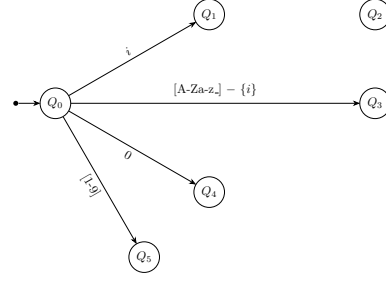
line	operation	value
13	if $T' \notin D$	$Q_4 \notin D$
14	$W := W \cup \{T'\}$	$W = \{Q_1, Q_3, Q_4\}$
15	$D := D'$	$D = \{Q_0, Q_1, Q_3, Q_4\}$
6	for $\alpha \in \Sigma$	$\alpha \in [1-9]$
7	$T := \emptyset$	$T = \emptyset$
8	for $S \in Q$	$S = q_0$
9	$T := T \cup \delta_N(S, \alpha)$	$\emptyset \rightarrow \emptyset$
8	for $S \in Q$	$S = q_1$
9	$T := T \cup \delta_N(S, \alpha)$	$\emptyset \rightarrow \emptyset$
8	for $S \in Q$	$S = q_4$
9	$T := T \cup \delta_N(S, \alpha)$	$\emptyset \rightarrow \emptyset$
8	for $S \in Q$	$S = q_6$
9	$T := T \cup \delta_N(S, \alpha)$	$\emptyset \rightarrow \{q_8\}$
10	$T' := \varepsilon\text{-closure}(T)$	$T' = \{q_8\} = Q_5$
11	$D' := D \cup \{T'\}$	$D' = \{Q_0, Q_1, Q_3, Q_4, Q_5\}$



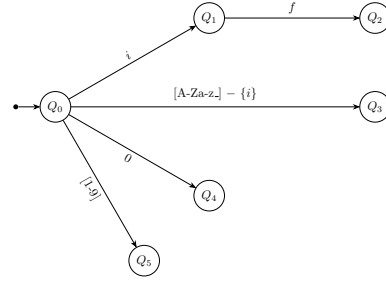
line	operation	value
12	$\delta_D(Q, \alpha) := T'$	$\delta_D(Q_0, \alpha) = Q_5$



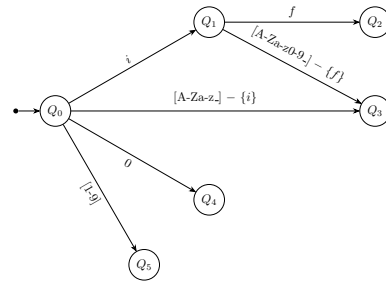
line	operation	value
13	if $T' \notin D$	$Q_5 \notin D$
14	$W := W \cup \{T'\}$	$W = \{Q_1, Q_3, Q_4, Q_5\}$
15	$D := D'$	$D = \{Q_0, Q_1, Q_3, Q_4, Q_5\}$
4	while $W \neq \emptyset$	$W = \{Q_1, Q_3, Q_4, Q_5\} \neq \emptyset$
5	remove Q from W	$Q = Q_1, W = \{Q_3, Q_4, Q_5\}$
6	for $\alpha \in \Sigma$	$\alpha = f$
7	$T := \emptyset$	$T = \emptyset$
8	for $S \in Q$	$S = q_2$
9	$T := T \cup \delta_N(S, \alpha)$	$\emptyset \rightarrow \{q_3\}$
8	for $S \in Q$	$S = q_5$
9	$T := T \cup \delta_N(S, \alpha)$	$\{q_3\} \rightarrow \{q_3, q_5\}$
10	$T' := \varepsilon\text{-closure}(T)$	$T' = \{q_3, q_5\} = Q_2$
11	$D' := D \cup \{T'\}$	$D' = \{Q_0, Q_1, Q_2, Q_3, Q_4, Q_5\}$



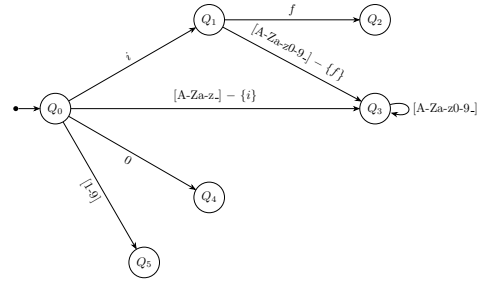
line	operation	value
12	$\delta_D(Q, \alpha) := T'$	$\delta_D(Q_1, \alpha) = Q_2$



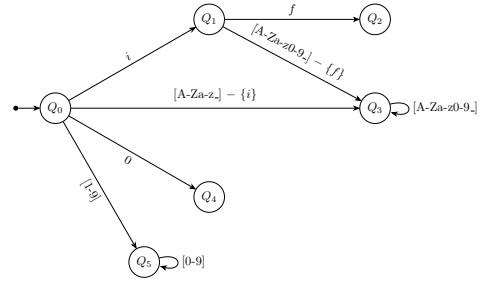
line	operation	value
13	if $T' \notin D$	$Q_2 \notin D$
14	$W := W \cup \{T'\}$	$W = \{Q_3, Q_4, Q_5, Q_2\}$
15	$D := D'$	$D = \{Q_0, Q_1, Q_2, Q_3, Q_4, Q_5\}$
6	for $\alpha \in \Sigma$	$\alpha \in [A-Za-z0-9] - \{f\}$
7	$T := \emptyset$	$T = \emptyset$
8	for $S \in Q$	$S = q_2$
9	$T := T \cup \delta_N(S, \alpha)$	$\emptyset \rightarrow \emptyset$
8	for $S \in Q$	$S = q_5$
9	$T := T \cup \delta_N(S, \alpha)$	$\emptyset \rightarrow \{q_5\}$
10	$T' := \varepsilon\text{-closure}(T)$	$T' = \{q_5\} = Q_3$
11	$D' := D \cup \{T'\}$	$D' = \{Q_0, Q_1, Q_2, Q_3, Q_4, Q_5\}$
12	$\delta_D(Q, \alpha) := T'$	$\delta_D(Q_1, \alpha) = Q_3$



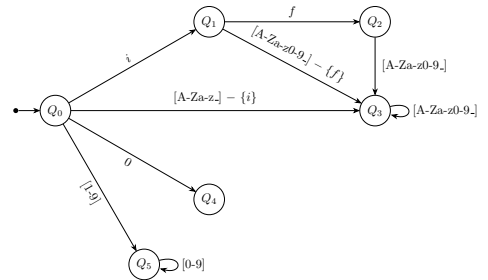
line	operation	value
13	if $T' \notin D$	$Q_3 \in D$
15	$D := D'$	$D = \{Q_0, Q_1, Q_2, Q_3, Q_4, Q_5\}$
4	while $W \neq \emptyset$	$W = \{Q_3, Q_4, Q_5, Q_2\} \neq \emptyset$
5	remove Q from W	$Q = Q_3, \quad W = \{Q_4, Q_5, Q_2\}$
6	for $\alpha \in \Sigma$	$\alpha \in [A-Za-z0-9_]$
7	$T := \emptyset$	$T = \emptyset$
8	for $S \in Q$	$S = q_5$
9	$T := T \cup \delta_N(S, \alpha)$	$\emptyset \rightarrow \{q_5\}$
10	$T' := \varepsilon\text{-closure}(T)$	$T' = \{q_5\} = Q_3$
11	$D' := D \cup \{T'\}$	$D' = \{Q_0, Q_1, Q_2, Q_3, Q_4, Q_5\}$
12	$\delta_D(Q, \alpha) := T'$	$\delta_D(Q_3, \alpha) = Q_3$



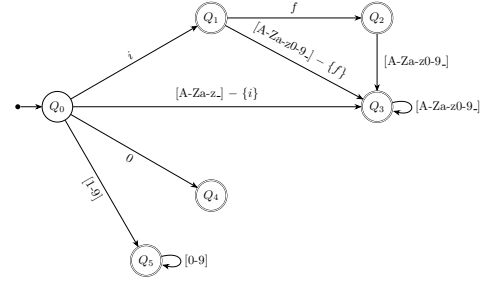
line	operation	value
13	if $T' \notin D$	$Q_3 \in D$
15	$D := D'$	$D = \{Q_0, Q_1, Q_2, Q_3, Q_4, Q_5\}$
4	while $W \neq \emptyset$	$W = \{Q_4, Q_5, Q_2\} \neq \emptyset$
5	remove Q from W	$Q = Q_4, \quad W = \{Q_5, Q_2\}$
6	for $\alpha \in \Sigma$	$\alpha \in [A-Za-z0-9_]$
7	$T := \emptyset$	$T = \emptyset$
8	for $S \in Q$	$S = q_7$
9	$T := T \cup \delta_N(S, \alpha)$	$\emptyset \rightarrow \emptyset$
10	$T' := \varepsilon\text{-closure}(T)$	$T' = \emptyset, \quad \delta_D(Q_4, \alpha) = \text{undefined}$
4	while $W \neq \emptyset$	$W = \{Q_5, Q_2\} \neq \emptyset$
5	remove Q from W	$Q = Q_5, \quad W = \{Q_2\}$
6	for $\alpha \in \Sigma$	$\alpha \in [A-Za-z_]$
7	$T := \emptyset$	$T = \emptyset$
8	for $S \in Q$	$S = q_8$
9	$T := T \cup \delta_N(S, \alpha)$	$\emptyset \rightarrow \emptyset$
10	$T' := \varepsilon\text{-closure}(T)$	$T' = \emptyset, \quad \delta_D(Q_5, \alpha) = \text{undefined}$
6	for $\alpha \in \Sigma$	$\alpha \in [0-9]$
7	$T := \emptyset$	$T = \emptyset$
8	for $S \in Q$	$S = q_8$
9	$T := T \cup \delta_N(S, \alpha)$	$\emptyset \rightarrow \{q_8\}$
10	$T' := \varepsilon\text{-closure}(T)$	$T' = \{q_8\} = Q_5$
11	$D' := D \cup \{T'\}$	$D' = \{Q_0, Q_1, Q_2, Q_3, Q_4, Q_5\}$
12	$\delta_D(Q, \alpha) := T'$	$\delta_D(Q_5, \alpha) = Q_5$



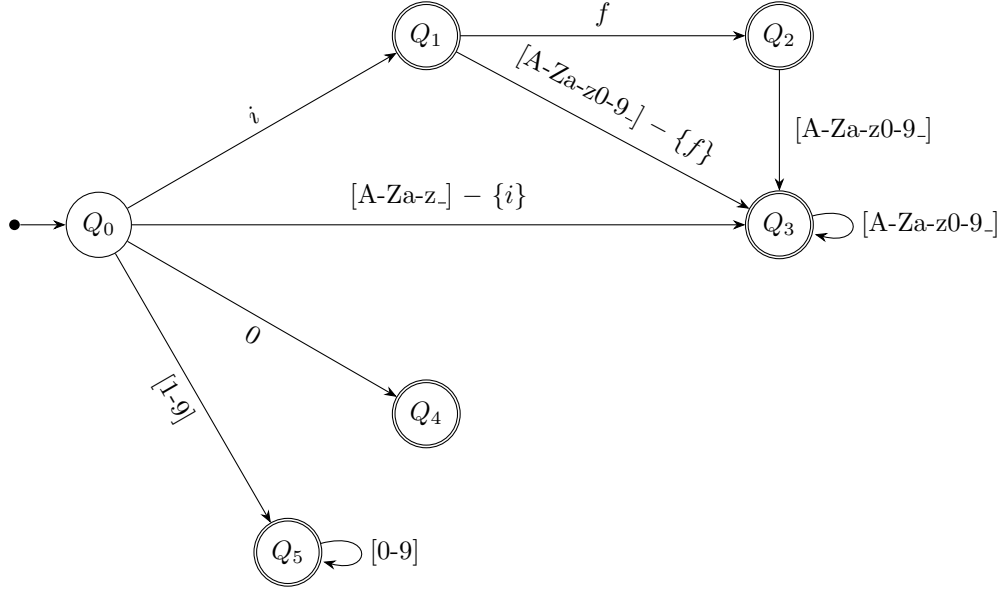
line	operation	value
13	if $T' \notin D$	$Q_5 \in D$
15	$D := D'$	$D = \{Q_0, Q_1, Q_2, Q_3, Q_4, Q_5\}$
4	while $W \neq \emptyset$	$W = \{Q_2\} \neq \emptyset$
5	remove Q from W	$Q = Q_2, \quad W = \{\}$
6	for $\alpha \in \Sigma$	$\alpha \in [A-Za-z0-9_]$
7	$T := \emptyset$	$T = \emptyset$
8	for $S \in Q$	$S = q_3$
9	$T := T \cup \delta_N(S, \alpha)$	$\emptyset \rightarrow \emptyset$
8	for $S \in Q$	$S = q_5$
9	$T := T \cup \delta_N(S, \alpha)$	$\emptyset \rightarrow \{q_5\}$
10	$T' := \varepsilon\text{-closure}(T)$	$T' = \{q_5\} = Q_3$
11	$D' := D \cup \{T'\}$	$D' = \{Q_0, Q_1, Q_2, Q_3, Q_4, Q_5\}$
12	$\delta_D(Q, \alpha) := T'$	$\delta_D(Q_2, \alpha) = Q_3$



line	operation	value
13	if $T' \notin D$	$Q_3 \in D$
15	$D := D'$	$D = \{Q_0, Q_1, Q_2, Q_3, Q_4, Q_5\}$
4	while $W \neq \emptyset$	$W = \{\}$
16	$F.D := \{Q \in D \mid Q \cap F.N \neq \emptyset\}$	$F.D = \{Q_1, Q_2, Q_3, Q_4, Q_5\}$



2.3 DFA



A Appendix: Subset Construction Algorithm

line	operation
1	$D0 := \varepsilon\text{-closure}(\{N0\})$
2	$D := \{D0\}$
3	$W := \{D0\}$ // worklist
4	while $W \neq \emptyset$:
5	remove Q from W
6	for $\alpha \in \Sigma$:
7	$T := \emptyset$
8	for $S \in Q$:
9	$T := T \cup \delta_N(S, \alpha)$
10	$T' := \varepsilon\text{-closure}(T)$
11	$D' := D \cup \{T'\}$
12	$\delta_D(Q, \alpha) = T'$ // update δ_D
13	if $T' \notin D$:
14	$W := W \cup \{T'\}$
15	$D := D'$
16	$F.D := \{Q \in D \mid Q \cap F.N \neq \emptyset\}$ // define $F.D$